Did Humans Invent Language?

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Abstract

No non-human animal has spontaneously acquired language with open-ended vocabulary and rule-governed syntax. This paper proposes that language was invented by early modern *Homo sapiens*—a cultural act by cognitively capable individuals. Three clusters of novel hypotheses follow from this invention thesis. First, the language invention cluster: the Evolutionary Barrier to Language Hypothesis posits that no species has evolved language by incremental natural selection alone; the Language Invention Hypothesis proposes that humans invented a Free-Form Protolanguage (FFP) of symbolic words and flexible word combinations without grammatical structure—sufficient, together with shared context, to convey whole thoughts; and the Two-Stage Language Origin Hypothesis holds that this culturally invented FFP preceded and drove the biological evolution of recursive syntax. Second, the information-theoretic consequences of invention: the Natural Language Codec Hypothesis proposes that vocabulary and grammar function as compression algorithms, producing an effective data rate (EDR) of spoken language far exceeding the acoustic channel rate of ~39 bits/s (Coupé et al., 2019). Third, the biological consequences of invention: the Rapid Language Diffusion Hypothesis posits that language provided decisive competitive advantages, driving global dispersal of *H. sapiens*; the Language Lifespan Extension Hypothesis proposes that language extended survival into old age, generating two subsidiary hypotheses: the Grandparenting Hypothesis, in which knowledge transmission created selection for elder survivorship in both sexes, and the Fossil Menopause Hypothesis, in which menopause is a remnant of an ancestral limit. Together, these hypotheses suggest that language was the most consequential invention in human history.

Introduction

Most work on language origins treats language as a product of incremental evolution by natural selection. This view presents a difficulty: no other species has evolved open-ended symbolic language, despite millions of years of opportunity. Language appears to have evolved only once—suggesting that selection pressure alone was insufficient. The evolution of recursive syntax has been called an unsolved mystery (Hauser et al., 2014).

An alternative is that language was invented, and that the biological evolution of syntax followed rather than preceded it. Everett (2017) anticipated the idea that language was a creation of culture; Bickerton (1990) and Tomasello (2008) proposed protolanguage theories in which words preceded grammar. Dor and Jablonka (2001) modelled how culturally transmitted linguistic innovation could drive genetic assimilation of the capacity to acquire language, but explicitly set aside the question of origin. That question is the subject of this paper, together with its information-theoretic consequences.

This paper proposes a novel interpretation of existing evidence: eight interlinked hypotheses that together describe a single sequence—an evolutionary barrier, its breach, and a cascade of genetic, life-history, and cultural consequences. Early modern humans crossed that barrier by inventing language.

1 Animal Communication

No non-human animal has acquired language with open-ended vocabulary and rule-governed syntax (Hage, 2024). Vervet monkeys have largely innate alarm calls specific for leopards, eagles, and snakes (Seyfarth et al., 1980). Wild bonobos and chimpanzees combine pairs of innate calls (e.g., yelp-grunt) into structures with modified meanings (Berthet et al., 2025; Girard-Buttoz et al., 2025).

Vocal production learning—the ability to imitate or invent sounds—is rare (Janik & Slater, 1997). Wild chimpanzees do not mimic sounds and do not invent new symbolic vocalisations. Attempts to teach chimpanzees spoken language (Hayes & Hayes, 1951) and sign language produced limited results—apes learned signs but showed no evidence of productive syntax or spontaneous symbol creation (Gardner & Gardner, 1969; Terrace et al., 1979). With intensive training, the bonobo Kanzi used ~400 lexigrams and reportedly understood ~3,000 spoken English words (Savage-Rumbaugh & Lewin, 1994), yet showed no evidence of recursive syntax or open-ended word coinage. Critically, Kanzi could not create new symbols to represent concepts for which he lacked lexigrams. When humans teach them, wild animals can sometimes learn a symbolic system with syntax—but none has invented one.

Evolutionary barriers range from low to insuperable. Eyes independently evolved more than 40 times. Powered flight evolved only four times. Non-avian dinosaurs did not evolve language in over 160 million years. In the 66 million years since the K–Pg extinction event, no mammal lineage evolved language until *H. sapiens*.

The evolution of language by incremental natural selection alone may be impossible due to a fitness valley. Symbolic language requires several capacities—abstraction, shared convention, cooperative motivation, and displaced reference—whose benefits in isolation may have been insufficient to sustain incremental evolution. The Evolutionary Barrier to Language Hypothesis is summarised in Section 15.

2 Is Language Needed for Thought?

Chomsky proposed that language evolved as a "tool for thought" (Berwick & Chomsky, 2016). He maintained that "externalization" (vocal speech) evolved later and was secondary to internal cognition. Other linguists argue that language is primarily a tool for external communication (Fedorenko et al., 2024).

Many forms of conscious thought do not require language. The earliest stone tools date to approximately 2.6 million years ago (Klein, 2000). The large brains may have evolved before language to process and remember environmental knowledge, and to facilitate tool-making and organised hunting with weapons (Stout & Chaminade, 2012). Some argue that the deep structures of human cognition are not linguistic (Fedorenko et al., 2024) and that language primarily functions to provide a posteriori explanation of concepts to ourselves and to others.

Language serves to organise, communicate, and preserve ideas across generations, rather than leaving them locked inside the heads of isolated individuals. Language enabled cumulative culture: a ratchet effect in which innovations are preserved, transmitted, and built upon across generations (Tomasello, 2008). Language may explain the accelerated technological change documented in the Upper Palaeolithic record.

3 Early Hominin Communication

There is ongoing debate about when the first elements of language appeared. The most common estimates for the emergence of a protolanguage fall between about 100,000 and 80,000 years ago. The evidence for early protolanguage includes:

- Brain size of *H. heidelbergensis* and Neanderthals similar to modern humans;
- Hyoid bone shape in *H. heidelbergensis* and Neanderthals similar to humans;
- Neanderthals had a FOXP2 gene variant (needed for fluent speech) identical to that of humans.

None of the above is definitive evidence of pre-language. Brain organisation and shape may be more important than volume. The appearance of functional human language may be marked in the archaeological record by a combination of characteristics:

- prolonged childhood and development
- increased survivorship into older adulthood
- high rate of technological innovation
- symbolic complexity and behavioural modernity
- long-distance trade and exchange networks
- rapid population growth

- outward migration
- rapid extinction of other *Homo* species

Recent evidence that Neanderthal infants experienced significantly faster somatic growth than modern human infants (Been et al., 2026) suggests that the slow early development characteristic of *H. sapiens* may be a specifically modern human adaptation. Neanderthal children did not grow abnormally fast; rather, *H. sapiens* children grow abnormally slowly. Humans spend almost twice as long in childhood and adolescence as chimpanzees and gorillas. This prolonged developmental timeline (Locke & Bogin, 2006) is consistent with the extended critical period for language acquisition (Lenneberg, 1967). In addition, the ratio of older to younger adults rose roughly fivefold at the Upper Palaeolithic transition (Caspari & Lee, 2004), also consistent with language acquisition.

H. heidelbergensis and Neanderthals had large brains but low rates of technological innovation, with their toolkits remaining largely unchanged for some 200,000 years (Klein, 2000). Simple tool-making—knapping an Acheulean hand axe, for instance—can be learned by observation and practice, without language (Stout & Chaminade, 2012). Symbolic cultural behaviour in *H. sapiens* appeared at least 77,000 years ago in Africa with shell beads and engraved ochre (Henshilwood et al., 2002). This sudden acceleration in innovation may indicate the earliest effects of language on human culture.

Some linguists think that human language originated only once (Atkinson, 2011). Chomsky argued that the core computational property of language (Merge) arose from a single chance mutation in one individual within an ancestral *H. sapiens* population, subsequently spreading through the group (Berwick & Chomsky, 2016)—a disputed saltationist view of the evolution of syntax. Since all humans have equal linguistic capability, language likely originated in an ancestral African population before regional diffusion and cultural diversification (Lenneberg, 1967), consistent with a single origin of languages.

After about 45,000 years ago, modern humans in Europe progressed through a rapid succession of distinct tool traditions. The Upper Palaeolithic Revolution (~50,000–10,000 years ago) saw an explosion of sophisticated tools, including stone blades, composite hafted tools, atlatl spear throwers, and cave paintings (Klein, 2000). This cultural flowering may have resulted from the cumulative social effects of language.

The population of *H. sapiens* expanded in Africa 80,000–60,000 years ago (Mellars, 2006) *and* out of Africa about 60,000–45,000 years ago (Henn et al., 2012), primarily via a coastal migration route (Macaulay et al., 2005), though multiple dispersal events are also supported by genomic evidence. All other *Homo* species went extinct, perhaps unable to compete with *H. sapiens*.

Taken together, the archaeological evidence points to the advent of fully modern, fully syntactic language around 80,000–50,000 years ago—coincident with the demographic expansion and the disappearance of other *Homo* species. However, correlation does not prove causation.

4 Precursors for Language

Several biological and cognitive preconditions appear to have been necessary before language could be invented. Intentional gestural communication—pointing, iconic gestures, and facial expressions—may have established the joint attention and referential intent that symbolic words later exploited (Corballis, 2002; Tomasello, 2008). While gesture may have seeded referential intent, its limited bandwidth—a small, slowly produced sign set—is insufficient to carry a full symbolic vocabulary. Vocal production learning—the ability to produce and modify novel sounds—was equally important (Janik & Slater, 1997). These capacities, while necessary, are insufficient alone; many animal species share them without inventing language.

Great apes, our closest relatives, possess relevant cognitive capacities—joint attention, some symbolic comprehension under training (Savage-Rumbaugh & Lewin, 1994)—yet do not spontaneously invent or conventionalise a shared symbolic system. Great apes have never developed a protolanguage.

Musicality—the biological capacity to perceive and respond to structured sound—may have been the most important precondition. Musicality arises from pre-existing auditory and cognitive capacities (Trainor, 2015), is partially shared with other vocal learning species (Patel, 2021), and involves neural circuits partially overlapping with those used for language (Peretz et al., 2015; Honing, 2026). Music-making—the culturally transmitted production of structured sound—appears to be unique and universal to humans (Honing, 2026; Fitch, 2006).

Invented music and invented words share critical properties: both require mapping arbitrary sounds to learned meanings, and both demand the fine voluntary vocal control that distinguishes *H. sapiens* from other primates. Musicality and language may share a common neural architecture that evolved independently of general intelligence (Banerjee et al., 2026). Musicality may have been the biological scaffold on which the cultural invention of music, and ultimately language, was built.

These preconditions—joint attention, vocal production learning, and musicality—were necessary but not sufficient. Also needed was the creative capacity for metaphorical abstraction required to map arbitrary sounds to meanings. Cognitive modernity, present in *H. sapiens*

~100,000 years ago, made the invention of language possible. What remained was the cultural act of invention itself.

5 Was Language Invented?

The Language Invention Hypothesis posits that a small group of early modern humans may have had the capacity to invent words with assigned definitions. Humans can form mental images from memory or imagination, the cognitive basis for displacement. A single individual could have coined many words for common objects and actions, attaching abstract meaning to arbitrary spoken sounds. Such acts occur within a single lifetime, not the millennia required for biological evolution. This innovation was a novel form of coded, symbolic thought not seen previously in nature.

The invention of language may not have succeeded at first but may have occurred more than once. With a successful attempt, utilitarian words could have been copied and added to by others—a local cultural innovation and the beginning of language as a shared skill carried beyond the lifetime of its inventors. The propagation of a linguistic innovation may have been the real bottleneck (Dor & Jablonka, 2001).

Structured syntax was not essential for an early, rudimentary protolanguage. Unstructured word groups can convey information and intentions effectively (e.g., *gazelle—valley—you—spear—kill*). In context, non-compositional groups of words with flexible order often work well enough. This invented Free-Form Protolanguage (FFP) may have provided the first means to express complete ideas.

The FFP may have been facilitated by the human capacity for joint attention and eye gaze following (Tomasello, 2008), enabling early language users to establish shared context. While the symbolic vocabulary was invented, the FFP likely inherited the paralinguistic features—pitch, stress, tempo, intonation—already present in ancestral primate vocalisation. These carried part of the load grammar would later assume. Embedded within spoken words, such cues supplemented situational context.

The FFP was a learned, symbolic, referential communication system—a language without regular syntax. FFP words were genuinely symbolic rather than indexical: a word such as *bird* denotes an abstract category, and *run* applies to any agent capable of the action. Their meanings are defined by relationships among symbols and concepts, rather than by correlation with a current moment. Crucially, FFP words exhibit displacement—they can be used in the absence of their referents.

The concept of a protolanguage was anticipated by earlier theorists. Bickerton (1990) argued that early humans gradually evolved a protolanguage with isolated words and simple phrases but no grammar. Tomasello (2008) similarly argued that words and intentional communication preceded grammatical structure, attributing this sequence to the gradual evolution of shared intentionality. The FFP hypothesis differs from both: it proposes that protolanguage was a de novo cultural invention by cognitively modern *H. sapiens*, not a relic of biological evolution in earlier hominins.

Iterated-learning research shows that combinatorial, compositional structure can emerge cumulatively as a language is transmitted across generations, without intentional design (Kirby et al., 2008). Kirby addresses the origins of linguistic structure rather than the origin of symbolic reference—the latter being the invention hypothesis proposed here.

An FFP could have been widely adopted and elaborated across many generations, with an ever-expanding vocabulary and words concatenated in any order. Even a rudimentary FFP would have provided survival advantages during the uniquely difficult and dangerous process of human childbirth (Trevathan, 2011). The FFP resembled "telegraphic speech"—used by children and adults in high-concentration tasks. Verb-noun compounds such as *scarecrow* have been identified as "living fossils" of simpler, earlier language structures (Jackendoff, 2002; Progovac, 2015), consistent with an FFP.

The FFP was functional but limited without syntax. Lacking grammatical structure, it could not always specify who-did-what-to-whom, so meaning relied on shared context. Complex expressions would have required ever-larger word groups, a combinatorial burden. Without syntax, the FFP lacked the expressive power and efficiency of modern language.

6 The Evolution of Syntax

The invention of the FFP may have been a necessary precondition for language to fully evolve. Once the FFP was in use, it likely created strong selective pressure for regularised, high-speed grammar. This second stage of language development may have required the incremental evolution of specialised neural circuitry for fluent speech (Progovac, 2015). Deacon (1997) argued that symbolic language use created selective pressure for the co-evolution of the prefrontal cortex—language pulling its own biology into existence.

The invention of the FFP constituted a niche construction event (Odling-Smee et al., 2003) of unprecedented scope. Profound human inventions can trigger cascading evolutionary change. The invention of tools and cooking with fire may have remodelled human digestive anatomy and facial morphology (Wrangham, 1999). In a similar fashion, language created a self-modifying cultural environment that transformed the selective pressures acting on human populations, initiating a process of culture-gene coevolution.

The biological evolution of language was accompanied by further cascading changes. The descended larynx enabled richer phonemic production, at the high cost of increased choking risk. A prolonged juvenile period evolved to accommodate cultural knowledge and language acquisition, at the high cost of extended dependency. The uniquely flat, gracile face of *H. sapiens*—in contrast to the projecting mid-face and heavy brow ridges of Neanderthals and earlier hominins—may reflect selection for both the fine articulatory movements of speech and the paralinguistic facial expressions that enrich verbal communication (Godinho et al., 2018). Uttley et al. (2025) showed that modern humans carry a less active version of a non-coding SOX9 enhancer than Neanderthals, consistent with our more gracile mandibles and potentially contributing to the vocal tract reconfiguration for fluent speech. Refinement of precision hand grip morphology in *H. sapiens*—compared to the power-grip-optimised hands of Neanderthals—supported both fine tool use and the gestural communication that accompanied language (Greenfield, 1991).

The FFP existed in a social context. Clever, fluent speech was attractive, and sexual selection likely provided a secondary pressure reinforcing the primary functional demand for grammar. Progovac (2026) proposed that sexual selection for quick-wittedness and humour played a causal role from the earliest stages of human language evolution, and Darwin (1871) proposed that eloquence—verbal ornamentation signalling cognitive fitness—drove brain evolution for spoken language. Even today, well-spoken individuals enjoy social and mate-selection advantages.

The **Two-Stage Language Origin Hypothesis**—the cultural invention of a free-form protolanguage followed by biological evolution of specialised neural circuitry for recursive syntax—may explain why open-ended language with syntax did not evolve elsewhere in nature.

The required sequence of cultural invention *preceding* biological selection—once symbols were invented, syntax could evolve—may explain how human creativity broke through the high evolutionary barrier to language. The two stages are illustrated in Fig. 1.

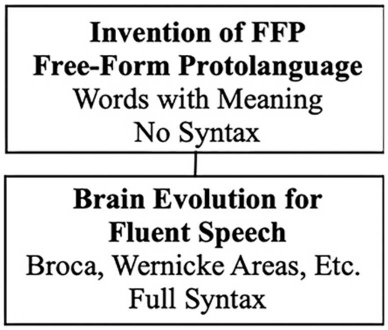

Fig. 1 Two-Stage Language Origin Hypothesis: cultural invention of a protolanguage followed by biological evolution of neural circuitry for syntactic speech.

The stages of the Two-Stage Language Origin Hypothesis operated on different timescales. The acquisition of language was at first rapid and then gradual: a sudden creative act overcame the evolutionary barrier that selection alone could not cross, thereafter freeing slower natural selection to perfect the neural machinery for recursive syntax.

Stage 1—the cultural invention of an FFP—could have become established within a few generations, around 100,000–80,000 years ago, consistent with the earliest surviving symbolic artefacts. Engraved ochre at Blombos Cave (~77,000 years ago; Henshilwood et al., 2002) shows a capacity for arbitrary, conventional symbolism—the faculty an FFP requires. Because such artefacts index symbolic reference rather than grammar, they bear on Stage 1.

Stage 2—the evolutionary optimisation of neural circuitry for recursive syntax—would have started soon after the FFP was established and unfolded over a much longer period.

Because the FFP could not reliably convey complex or novel propositions without syntax, its use generated persistent directional selection for neural machinery supporting hierarchical syntactic structure. Stage 2 would have recruited and refined pre-existing neural variation (Dor & Jablonka, 2001). Consistent with this, Jackendoff and Pinker (2005) argue that recursion is a novel combination of capacities partly present elsewhere in cognition.

The African archaeological record provides an empirical constraint: the roughly 20,000–30,000-year interval between the appearance of unambiguous symbolic artefacts (~80,000–70,000 years ago) and the Upper Palaeolithic cultural explosion (~50,000 years ago) represents approximately 800–1,200 human generations, assuming a generation time of 25 years. This 800–1,200-generation window may have been sufficient for the development of neural circuitry for recursive syntax.

A key implication of the Two-Stage Language Origin Hypothesis is that Stage 1 required cognitive modernity but not modern brain shape—the FFP could be invented with the neural architecture already present in early *H. sapiens* by ~300,000 years ago (Neubauer et al., 2018). Genomic analysis independently indicates that linguistic capacity was present by at least ~135,000 years ago (Miyagawa et al., 2025), consistent with the substrate for language predating its invention.

The globularisation of the cerebellum and parietal lobe—structures central to language processing and articulation—was completed gradually between ~100,000 and ~35,000 years ago (Neubauer et al., 2018), coinciding with the proposed timing of Stage 2. Multiple factors may have contributed, including dietary change and expanded social complexity. Language is another plausible explanation that predicts incremental selection for neural circuitry directly implicated in language processing. Hemispheric lateralisation predates language; handedness and cortical asymmetries may have been exapted for syntactic processing in Stage 2, with sustained selection intensifying left-hemisphere language dominance.

Generalised grammatical speech requires specialised neural structures, including Broca's and Wernicke's areas. These metabolically costly structures are universal in modern humans. The proposed culture-gene feedback loop—FFP use favouring better neural machinery, which enabled richer syntax, which then drove further brain evolution—may have been self-amplifying, compressing the Stage 2 timeframe.

7 Natural Language Codec Hypothesis

The invention of words for the FFP required the cognitive capacity for metaphorical abstraction (Lakoff & Johnson, 1980). De novo word coinage maps a concrete reality onto an arbitrary sound, compressing associated meaning into a single symbolic token—coded, symbolic thought. Invented language is thus not merely a labelling system but a novel Natural Language Codec (NLC) communication system.

Codecs are essential for efficient information transfer because they compress and structure information, increasing the effective data rate of communication (MP3, for example). Encoding and decoding are separate operations, which may occur at different locations or times. Codecs can be lossless (no data lost) or lossy (some data lost to achieve greater compression), and are characterised by a compression ratio (CR) (Watkinson, 2002).

Analogous biological codec systems are also found in nature: the immune system encodes information about past pathogens in memory B cells, later decoded to manufacture neutralising antibodies.

Human speech and perception operate as a codec system—the speaker compresses meaning into sound, the listener decompresses sound into meaning, using shared vocabulary and grammar as the codec algorithms. NLCs overcome the inherent information-rate limits of the speech channel: Shannon (1948) established the information-theoretic framework later used to estimate the speech information rate (IR). Coupé et al. (2019) used syllabic information density and speech rate to measure a consistent IR of ~39 bits/s across 17 languages from 9 language families. However, speech information rate does not account for lexical meaning (Crocker et al., 2016) or semantic content.

Words carry symbolic meanings that transcend the acoustic limit: the effective data rate (EDR) is the rate of decoded semantic content, which exceeds the acoustic information rate by the codec's compression ratio (EDR = IR × CR). Because both the information rate and the compression ratio are defined per syllable, they combine multiplicatively to yield the semantic data rate. Both IR and EDR are expressed in bits/s, but they measure different quantities—channel throughput versus encoded or decoded meaning—so their ratio, not their difference, is what matters. Semantic expansion estimates should correlate with comprehension latency, word frequency, concreteness, and contextual predictability.

Both individual words and word groups constituted novel codecs. An FFP featured groups of words without fixed grammatical rules; this combinatorial productivity amplified semantic content, allowing early language users to express intentions and coordinate actions far beyond the capacity of individual signals. Unlike biological codecs, which evolved over millions of years, Natural Language Codecs were invented and refined culturally, at a pace far exceeding biological evolution (Perreault, 2012). This distinction may explain the sudden appearance of behaviourally modern humans in the archaeological record (Klein, 2000).

8 Lexical Codec

The initial inventive step toward symbolic language was the lexical codec. What seems obvious today, that words have meaning, was completely novel at the dawn of language. An arbitrary utterance was assigned an abstract definition, such as a name for a person, an object, or an action—the beginnings of a lexicon. For this innovation to succeed, a newly coined word had to be learned and understood by others, requiring innate vocal learning ability.

Humans use vocabulary to partially compensate for the low IR of the speech channel (Jackendoff, 2010). Memorised vocabulary contains compressed information about the meaning of words. The single-syllable word *horse* refers to a large four-legged animal; the superficially similar word *house* refers to a dwelling with a roof, windows, and a door. These seemingly simple observations form the foundation of the Lexical Codec.

Coupé et al. (2019) used the syllable as a unit for measuring the IR of various languages. Following that precedent, the syllable is used as the information-encoding unit for CR and EDR calculations. Phonemes can serve equally as the discrete unit; calculations using phonemes yield similar EDR values. What matters for the codec calculations is that vocalisation is divided into discrete, recombinable units (Jackendoff, 2002), whether syllables or phonemes.

The word *horse* has one syllable. A short dictionary definition requires 36 syllables: "a large plant-eating domesticated mammal with solid hooves and a flowing mane and tail, used for riding, racing, and to carry and pull loads"—a syllabic compression ratio of 36:1. Speaking time agrees with this: *horse* takes approximately 0.3 seconds; its definition takes approximately 11 seconds ($CR = 11 \div 0.3 \approx 36$).

The CR can be variable, according to circumstances and knowledge. For a person with real-world experience of horses—their temperament, mass, movement, and scent—the word conjures a far richer representation than a dictionary definition captures. If the listener does not know what a horse is, or a sparse definition is used—"a large hoofed mammal" (5 syllables)—then the CR and information communicated are both reduced. Semantic richness is thus only approximated by definition length. The example lexical EDR calculation below represents a conservative estimate.

The CR measures how many syllables of meaning are compressed into a single syllable. The effective data rate (EDR) is calculated by multiplying the speech channel IR by the CR:

EDR = IR × CR = 39 bits/s × 36 = 1,404 bits/s

Comprehension is rapid: the meaning of words is evoked with a neural latency of approximately 114 milliseconds (Brodbeck et al., 2018). Each word in the 36-syllable definition of horse is itself expandable with its own definition. A full accounting would include these nested definitions, so the first-order value of 1,404 bits/s is a conservative lower bound. Vocabulary words also encode additional information, such as a likely part of speech. The EDR example calculations here are restricted to first-order compression.

9 Syntactic Codec

The Syntactic Codec is an NLC that compresses the meaning of both FFP word groups and grammatical sentences. It is not a replacement for generative grammar or other theories of syntax. As with the Lexical Codec, the semantic gain of the Syntactic Codec is expressed as a compression ratio, quantifying the EDR achieved.

Just as the Lexical Codec requires learning thousands of words, the Syntactic Codec requires cultural knowledge of a specific language. Consider the sentence, “Einstein formulated relativity theory”—four words with a total of 13 syllables. When each word is unpacked using partial Wikipedia and dictionary definitions, Einstein is described using 277 words (506 syllables). Formulated is defined in 10 syllables. Relativity theory is defined in 59 words (135 syllables). The combined syllable count for all three definitions is 651. The CR is therefore 651:13, or 50:1.

As before, the EDR is calculated by multiplying the universal IR by the codec CR:

EDR = IR × CR = 39 bits/s × 50 = 1,950 bits/s

This is about 50 times the universal IR for human speech of ~39 bits/s (Coupé et al., 2019). The CR and EDR of the Syntactic Codec are instantaneous measures; both will vary from moment to moment, with language, speech rate, sentence complexity, and other factors.

Fig. 2 shows a conceptual flowchart of a general Syntactic Encoder Algorithm for an NLC. A speaker first conceives the thoughts and emotions to be expressed. Vocabulary words are then selected from a memorised lexicon, aided by context and the part-of-speech tag (e.g., noun, verb) associated with each word. The selected words are encoded and compressed to generate a syntactically correct sentence—a process that may embody Chomsky's Merge or some other mechanism of generative syntax. Finally, the sentence is spoken with the desired emotional tone, employing the approximately 100 coordinated muscles involved in fluent speech.

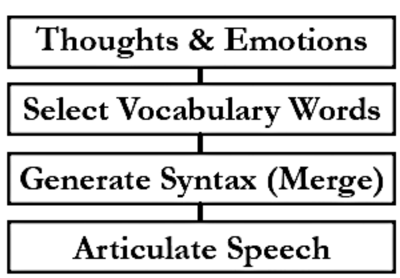

Fig. 2 Syntactic Encoder Algorithm: conceptual flowchart of the encoding process for a Natural Language Codec.

Chomsky's (1995) Minimalist Program posits that Merge is a fundamental syntactic operation combining two syntactic elements into a new hierarchical structure—this can be interpreted as an NLC compression algorithm. However, Merge describes only half of a Syntactic Codec; it specifies an encoder but not a decoder. Although Merge has been influential, alternative models of generative syntax exist (Friederici, 2011).

Fig. 3 shows a conceptual flowchart of a general Syntactic Decoder Algorithm for an NLC. The decoding process begins with hearing spoken words. The listener recognises each word from a memorised vocabulary, along with its part-of-speech tag. Drawing on a shared language, a predicted syntactic structure (Nakamura et al., 2026), and context, the listener can then parse the meaning of the sentence.

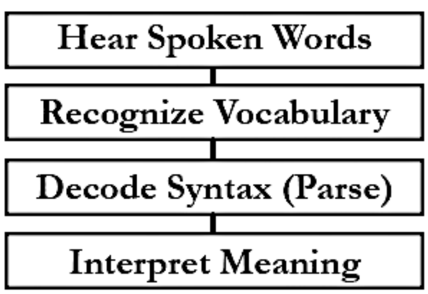

Fig. 3 Syntactic Decoder Algorithm: a conceptual flowchart of the decoding process for a Natural Language Codec.

Parsing syntax can begin rapidly, even for high-EDR sentences. When three-word English strings were briefly presented as a whole, the left temporal cortex distinguished subject–verb–object sentences from noun lists within about 130 milliseconds (Fallon & Pylkkänen, 2024). In ordinary speech, incremental, overlapping syntactic and semantic processing helps make fluent conversation possible.

10 Matching Codec Encoders and Decoders

Computer codecs usually have an identifying suffix—such as MP3 for audio, JPG for images—to ensure that the correct codec algorithm is used to compress, decode, and display the data. If data transmission errors occur, error correction routines can often recover corrupted or incomplete data.

NLCs use asymmetrical encoding and decoding processes and are specific to a particular language. Like computer codecs, the NLC encoder and decoder must match—the speaker and listener must share the same language. Native speakers can also predict sentence structure (Nakamura et al., 2026), which aids rapid comprehension. Most English speakers tolerate some variation and error, though misunderstandings do occur. This robustness reflects the efficiencies that shape human language, balancing clarity against effort (Gibson et al., 2019).

For conversational English, a speaker can adjust vocabulary and grammar to suit the listener—a teacher simplifying language for a young child, or a professional using specialised jargon with a peer. Beyond shared language, successful verbal communication often depends on shared cultural knowledge.

Native speakers of English can readily gauge another speaker's verbal competence. The voice conveys tone, mood, intention, and identity; a trained ear can further detect geographic origin, education, social class, and other social and cultural background markers.

11 Rapid Language Diffusion Hypothesis

The invention of the FFP may have driven rapid spread of language-speaking populations. When protolanguage first appeared between 100,000 and 80,000 years ago, the various African *Homo* populations were sparse and scattered (Hollfelder et al., 2021). However, there was a population explosion in Africa between 80,000 and 60,000 years ago (Mellars, 2006), when humans spread to diverse habitats (Hallett et al., 2025), perhaps due to the adaptive advantages provided by language.

Independent genomic evidence supports this rapid dispersal model: Y-chromosome analysis places a cluster of major non-African founder haplogroups in a narrow window of 47,000–52,000 years ago, consistent with rapid initial colonisation of Eurasia and Oceania following an out-of-Africa bottleneck (Karmin et al., 2015)—within the window proposed here for the advent of fully modern language.

An out-of-Africa dispersal ~60,000 years ago led to the permanent establishment of *H. sapiens* populations in Europe, followed by the disappearance of Neanderthals. Burke et al. (2026) provide independent spatial modelling evidence that *H. sapiens* social networks in Europe were substantially more connected than those of Neanderthals during the critical period of 60,000–35,000 years ago. Language is a plausible explanation for this connectivity advantage. Language also enabled humans to remember the past and plan for the future (Suddendorf & Corballis, 2007).

Some researchers argue that modern language was present in the common ancestor of humans and Neanderthals about 500,000 years ago and evolved gradually thereafter (Dediu & Levinson, 2013). This view relies on disputed interpretations of Neanderthal and genetic evidence. The present framework allows that precursor capacities, including the neural foundations of Stage 2, may be ancient while maintaining that language itself—an invented symbolic system—arose recently in *H. sapiens*.

After acquiring language, verbal humans may have outcompeted, extirpated, or absorbed the remaining African *Homo* populations prior to the major expansion out of Africa. Within *H. sapiens*, language may also have spread without displacement—a woman marrying into a non-speaking band (exogamy) could raise a cohort of fluent speakers within a single generation.

Recent genomic analysis (Miyagawa et al., 2025) argues that linguistic capacity in *H. sapiens* existed ~135,000 years ago and suggests that language catalysed modern human behaviour by about 100,000 years ago, consistent with the timeline proposed here. That account, however, treats linguistic capacity as a biologically evolved faculty, whereas the present paper argues that the FFP protolanguage was culturally invented. Miyagawa et al. also infer that because no living human population lacks language, linguistic capacity must have been present before ~135,000 years ago; the same observation is equally consistent with invented language arising later in a single ancestral population that subsequently replaced or absorbed non-speaking groups, including non-speaking *H. sapiens*.

The first group of early modern humans to achieve language likely discovered its overwhelming competitive advantages. No other species can long withstand human ecological dominance (Shea, 2023). Outside Africa, the extinction of Neanderthals and Denisovans—who remain only as fragments of DNA in the modern human genome—is strong evidence that *H. sapiens* expansion displaced them. These processes could explain why there is only one human species and why no human population lacks language.

A Demographic Estimate: An early FFP, without recursive syntax, may have provided significant competitive advantages. If so, it is possible to estimate the population growth of language speakers in Africa 80,000–60,000 years ago. The exponential population growth formula is:

$$x(t) = x_0 \cdot \left(1 + \frac{r}{100}\right)^t$$

where t is the number of years; x_0 is the initial population, and r is the percentage growth rate.

As a thought experiment, a small founding population (~10 people), with an exponential growth rate of 1% (doubling time of about 70 years, consistent with favourable conditions among modern hunter-gatherers), would have produced a total population of about 4,000 language speakers within 600 years. This is consistent with estimates that the ancestral African population numbered only a few thousand individuals around this period (Henn et al., 2012). The global ecological range that humans came to occupy could not have been achieved through biological evolution alone, which would require ~88 million years of divergence (Perreault, 2026). Cultural evolution must have been the enabling mechanism.

Proposed Language Timeline:

- ~300,000 years ago—modern brain *size* established; cognitive modernity present
- ~100,000–80,000 years ago—earliest symbolic artefacts; FFP invention proposed; brain globularisation begins
- ~80,000–50,000 years ago—syntactic language continues to emerge during the Upper Palaeolithic transition
- ~35,000 years ago—brain globularisation complete; fully modern brain shape

Taken together, the evidence points to the advent of fully modern, syntactically complex language around 80,000–50,000 years ago.

12 Did Language Change Human Life History?

Because humans use language constantly and effortlessly, it appears ordinary; yet its advent was extraordinary, conferring competitive advantages so decisive that it reshaped not only behaviour and distribution but human life history itself.

The life-history hypotheses proposed here—Lifespan Extension, Grandparenting, and Fossil Menopause—follow from this premise. They are offered as testable ideas rather than essential claims; the central thesis, that language was culturally invented, does not depend on them.

Language may have altered the biological parameters of human life history. Extensive FFP use would have created an early learning burden, and the selection that drove Stage 2 brain evolution likely extended the juvenile period in parallel. This account inverts an influential proposal by Locke and Bogin (2006), who argued that the uniquely human life-history stages of childhood and adolescence are preconditions for language. The framework proposed here reverses the arrow for the juvenile period: rather than extended childhood being required for language acquisition (Lenneberg, 1967), the prolonged juvenile period is proposed as partly a consequence of language use.

The invention of language not only drove a Palaeolithic cultural flowering (Klein, 2000) but may also have extended survivorship. Wild chimpanzees have a mean adult lifespan of approximately 30 years (Hill et al., 2001), and Neanderthals had few individuals living beyond 40 years (Trinkaus, 2011; Caspari & Lee, 2004). During the Upper Palaeolithic transition, about 50,000–30,000 years ago, the proportion of *H. sapiens* living to older adulthood (about twice reproductive maturity) increased roughly fivefold above that of Middle Palaeolithic hominids, including Neanderthals (Caspari & Lee, 2004). Language intensified the transmission of accumulated knowledge across generations, amplifying the value of elder experience (Kaplan et al., 2000).

A species-typical lifespan reflects a heritable life-history trade-off between high reproductive rate and longevity. Modern humans are outliers on this fast-slow continuum, with exceptional longevity relative to body size. The usual explanation is the Grandmother Hypothesis (Hawkes et al., 1998; Aimé et al., 2017), which postulates that post-reproductive females enhanced inclusive fitness by provisioning grandchildren, a self-reinforcing feedback loop promoting longer female lifespans.

The Language Lifespan Extension Hypothesis builds on this trade-off. Human longevity may be ancient, having evolved within a knowledge-intensive foraging niche (Kaplan et al., 2000) under the general life-history principle that low extrinsic mortality favours longer survival (Kirkwood, 1977).

The palaeodemographic record indicates that extended survivorship was recent and pronounced: the ratio of older to younger adults rose roughly fivefold at the Upper Palaeolithic transition (Caspari & Lee, 2004). This most plausibly reflects a rise in *realised survivorship*—more individuals living into old age as language-borne knowledge lowered adult mortality. Language altered how many elders were present, not the biological ceiling on human life.

Language may have amplified the value of longevity, increasing the fitness payoff of long-lived, knowledgeable elders of both sexes. The hypothesis is testable: the Upper Palaeolithic rise in elder survivorship should track with archaeological proxies for cumulative culture, long-distance exchange, and complex toolkits.

13 Grandparenting Hypothesis

The Language Lifespan Extension Hypothesis generates two subsidiary hypotheses. The Grandparenting Hypothesis generalises the Grandmother Hypothesis (Hawkes et al., 1998). The Grandmother Hypothesis invokes provisioning by post-reproductive females, but the Grandparenting Hypothesis identifies the transmission of survival knowledge through language by both sexes. This knowledge-based account builds on embodied-capital theory (Kaplan et al., 2000), which holds that human longevity was selected to repay investment in acquired skill.

The Grandparenting Hypothesis extends to males because knowledge, unlike food, can be shared without being divided. Caloric provisioning is finite: food given to one child is denied another, so it must be directed at known kin. Paternity uncertainty makes such targeting unreliable for men. Linguistic knowledge is not depletable—taught once, it benefits many at negligible cost, so precise relatedness matters far less. In small hunter-gatherer bands, where background relatedness is already high, an older man's impulse to teach what he knows yields fitness returns, even when he does not know or care which children or grandchildren carry his genes. Some of his kin will likely benefit.

Because language made the accumulated knowledge of long-lived individuals transmissible, post-reproductive grandmothers and grandfathers alike became repositories of practical knowledge. The provisioning benefits of grandmothering are real but likely represent a secondary, reinforcing feedback on that increase in survivorship. The cooperative breeding and childcare system described by Hrdy (2009) exemplifies the extended, multi-generational support that language facilitated.

The Grandparenting Hypothesis gains additional biological support from evidence that paternal caregiving in men is hormonally mediated. Storey et al. (2000) showed that expectant and new fathers experience hormonal shifts—rising prolactin, falling testosterone—that parallel those of mothers and prepare males for infant care. Hrdy (2024) argues that the capacity for primary caregiving is a latent trait in all human brains, an "alloparental substrate" awaiting activation by proximity to infants. Grandfathers who remained closely involved with grandchildren in ancestral societies would have been biologically primed for caregiving investment.

14 Fossil Menopause Hypothesis

Menopause is also a life-history event influenced by language. In most mammal species, reproductive lifespan equals somatic lifespan—females remain fertile until death. Wild great apes and Neanderthals had mean adult lifespans of approximately 30–40 years, so most females died well before age 50—the age at which the ovarian reserve is exhausted and menopause occurs in modern humans.

What distinguishes *H. sapiens* is the length of post-reproductive survival. The Fossil Menopause Hypothesis proposes that menopause is a fossil of an ancestral condition. Human female reproductive years are constrained by a finite ovarian reserve established at birth. Language-driven extension of survivorship pushed more women past this ancestral limit, exposing menopause as a constraint that had previously been invisible.

15 Discussion and Original Contributions

This paper proposes eight hypotheses about language origins and its biological consequences, spanning evolutionary theory, information science, and human life history. The eight hypotheses are:

- Evolutionary Barrier to Language Hypothesis

- Language Invention Hypothesis
- Two-Stage Language Origin Hypothesis
- Natural Language Codec Hypothesis
- Rapid Language Diffusion Hypothesis
- Language Lifespan Extension Hypothesis
- Grandparenting Hypothesis
- Fossil Menopause Hypothesis

Evolutionary Barrier to Language Hypothesis. In the history of life, no species—including *H. sapiens*—has spontaneously evolved language by incremental natural selection. If language were a product of natural selection acting on large, social brains, it should have evolved repeatedly. However, in the last 100,000 years, language has appeared only once. This paper proposes that language faces an evolutionary barrier: a fitness valley where incremental natural selection cannot produce arbitrary symbolic reference, because no incremental genetic path from animal signalling to symbolic reference passes through a continuous sequence of immediate benefits.

Language Invention Hypothesis. Humans overcame the evolutionary barrier by inventing a Free-Form Protolanguage (FFP). A single cognitively modern individual could have coined the first words and assembled simple groups of words—mapping arbitrary sounds to specific meanings. This bootstrapping process required no biological change. Prior protolanguage theories (Bickerton, 1990; Tomasello, 2008) attributed words-before-grammar to gradual biological evolution; this paper proposes deliberate cultural invention.

Two-Stage Language Origin Hypothesis. Everett (2017) argued that language was a cultural creation, and protolanguage theories placed words before grammar—but neither developed these insights into a causally explicit two-stage model. Stage 1 is the de novo cultural invention of a Free-Form Protolanguage (FFP) by cognitively modern *H. sapiens*: a combinatorial system demonstrating that semantic productivity does not require grammatical structure. The FFP was sufficient to convey whole thoughts and coordinate action. Because flexible word groups do not specify relations such as who-did-what-to-whom, the FFP relied on situational context and inference to recover intended meaning. The residual limitations of the FFP became the selection pressure that drove the evolution of syntax in Stage 2.

Stage 2 is the subsequent biological evolution of specialised neural circuitry for recursive syntax, driven by selection pressure from FFP use. This causally ordered sequence—cultural invention preceding biological evolution—may explain why open-ended language has not evolved elsewhere in nature: the conjunction of cognitive modernity and deliberate symbolic invention represents an evolutionary bottleneck that no other lineage has traversed.

Natural Language Codec Hypothesis. Invented language was not merely vocal abstraction—it was a novel codec communication system for compressed meaning. Prior information-theoretic analyses of spoken languages measure the information rate (IR) of the speech channel at ~39 bits/s (Coupé et al., 2019). This does not account for the semantic content unlocked by shared vocabulary and grammar (Crocker et al., 2016). This paper introduces Natural Language Codecs (NLCs) to quantify semantic compression. The Lexical Codec shows that individual words compress semantic content at effective data rates (EDRs) far exceeding the raw channel rate. The Syntactic Codec extends this to structured sentences, compressing meaning further still. Because NLCs were culturally invented rather than biologically evolved, they developed far faster than natural selection—a property that may explain the sudden appearance of behaviourally modern humans in the archaeological record.

Rapid Language Diffusion Hypothesis. Invented language provided decisive competitive advantages that no non-linguistic competitor could match. Genetic evidence indicates a great demographic expansion beginning approximately 80,000 years ago in Africa (Mellars, 2006) and accelerating with the dispersal out of Africa 60,000–45,000 years ago (Henn et al., 2012), coinciding with the Upper Palaeolithic cultural explosion and the advent of fully modern syntactic language.

Language Lifespan Extension Hypothesis. Human longevity evolved within a knowledge-intensive foraging niche (Kaplan et al., 2000). During the Upper Palaeolithic transition, realised human lifespans were extended, with the proportion of individuals surviving to ~30+ years increasing roughly fivefold (Caspari & Lee, 2004). This would imply a comparable increase in the number reaching old age (~60+ years)—if language-mediated cooperation also reduced mortality later in life. The Language Lifespan Extension Hypothesis applies to both sexes and generates two subsidiary consequences: the Grandparenting Hypothesis and the Fossil Menopause Hypothesis.

Grandparenting Hypothesis. The Grandmother Hypothesis (Hawkes et al., 1998) attributes extended female lifespan to provisioning by post-reproductive females—a real effect, but sex-specific, and driven by provisioning rather than the verbal knowledge transmission proposed here. This paper proposes a broader Grandparenting Hypothesis: language enabled the accumulation and transmission of survival knowledge across generations, creating selection for elder survivorship in both sexes. Because knowledge, unlike food, can be shared without being divided, this benefit does not depend on the certainty of paternity that provisioning models require. Post-reproductive individuals, grandmothers and grandfathers alike, became repositories of practical wisdom and expertise that only language could preserve and transmit. The Grandmother Hypothesis is subsumed within this larger framework as secondary feedback on an increase in elder survivorship whose primary driver was language.

Fossil Menopause Hypothesis. In most mammal species, reproductive lifespan equals somatic lifespan—females die after the fertile period. Menopause has typically been explained as an adaptation: post-reproductive grandmothers enhance inclusive fitness by provisioning grandchildren (Hawkes et al., 1998). This paper proposes the opposite—menopause is not an adaptation but a fossil, a remnant of an earlier era when few women survived long after fertility ceased. The fixed ovarian reserve established at birth is exhausted at around age 50. Language-borne knowledge lowered mortality, exposing a previously invisible reproductive constraint. Ovarian reserves did not evolve to match the increased survival of women into old age.

Chain of Reasoning: The eight hypotheses form one causal sequence: a barrier, a breach, and a cascade. Throughout evolutionary history, language never spontaneously evolved—an evolutionary barrier that selection alone could not cross (Evolutionary Barrier Hypothesis). Early modern humans crossed it not by evolving language but by inventing it (Language Invention Hypothesis). This first protolanguage—the free-form protolanguage (FFP)—was functional but limited, lacking syntax. It functioned as a lexical codec that compresses meaning beyond the acoustic channel (Natural Language Codec Hypothesis). Its selective advantages then drove the evolution of neural circuitry for recursive syntax (Two-Stage Language Origin Hypothesis). Now equipped with full syntactic language, speakers gained overwhelming advantages and diffused worldwide, displacing non-speaking populations (Rapid Language Diffusion Hypothesis). Because language makes cumulative survival knowledge transmissible across generations, the longevity of elders became selectively valuable. Reduced mortality increased survival into old age (Language Lifespan Extension Hypothesis), so that elders of both sexes could teach their descendants (Grandparenting Hypothesis). Female fertility did not extend to match this longer survival, leaving menopause a fossil of the pre-linguistic past (Fossil Menopause Hypothesis).

Summary Comments: Language origins cannot be directly observed. No experimental and observational evidence can reproduce the conditions of the first language-makers.

Each hypothesis is grounded in established evidence and makes predictions where possible. No non-human animal trained on symbolic communication should demonstrate de novo word coinage without human scaffolding. Brain globularisation should correlate with archaeological proxies for symbolic culture. Increased elder survivorship in *H. sapiens* should co-occur with cumulative cultural complexity in the fossil record.

Future research directions include quantitative testing of NLC compression ratios across languages and sentence types; genomic analysis of longevity-related alleles for signatures of selection in the Upper Palaeolithic; and, comparative studies of female post-reproductive lifespan across great ape species under natural versus protected conditions.

Together, these hypotheses suggest that language was perhaps the most consequential invention in human history. The invention of language was a cultural act that bootstrapped its own biology, extended human survivorship, and enabled the global dominance of a single human species within an evolutionary eyeblink.

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